

CMSC 207: Chapter 7 Homework

April 26, 2026

1. A function $F : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R} \times \mathbb{R}$ is defined as follows:

$$F(x, y) = (3y - 1, 1 - x) \text{ for all } (x, y) \in \mathbb{R} \times \mathbb{R}.$$

Prove that F is a one-to-one correspondence, that is, F is both one-to-one and onto.

Proof: F is one-to-one.

Suppose that $F(x_1, y_1) = F(x_2, y_2)$ for some (x_1, y_1) and (x_2, y_2) in $\mathbb{R}^2$.

Evaluating $F(x_1, y_1)$ gives:

$$F(x_1, y_1) = (3y_1 - 1, 1 - x_1).$$

Evaluating $F(x_2, y_2)$ gives:

$$F(x_2, y_2) = (3y_2 - 1, 1 - x_2).$$

Since $F(x_1, y_1) = F(x_2, y_2)$, we have:

$$(3y_1 - 1, 1 - x_1) = (3y_2 - 1, 1 - x_2).$$

For ordered pairs to be equal, their corresponding components must be equal. Thus:

$$3y_1 - 1 = 3y_2 - 1 \qquad 1 - x_1 = 1 - x_2.$$

Using elementary algebra, we find that $y_1 = y_2$ and $x_1 = x_2$. Therefore, $(x_1, y_1) = (x_2, y_2)$.

Since $F(x_1, y_1) = F(x_2, y_2)$ implies that $(x_1, y_1) = (x_2, y_2)$, F is one-to-one.

Proof: F is onto.

Let (a, b) be an arbitrary element of $\mathbb{R}^2$. Let $x = 1 - b$ and $y = (a + 1)/3$. It is trivial to see that x and y remain in $\mathbb{R}^2$.

Evaluating $F(x, y)$ gives:

$$F(x, y) = F\left(1 - b, \frac{a + 1}{3}\right) = \left(3\left(\frac{a + 1}{3}\right) - 1, 1 - (1 - b)\right) = (a, b).$$

Since $F(x, y) = (a, b)$, we have shown that (x, y) is the preimage of (a, b) under F . Thus, for any $(a, b) \in \mathbb{R}^2$, there exists a preimage (x, y) under F . Therefore, F is onto.

Since F is both one-to-one and onto, F is a one-to-one correspondence. ■

2. Let S be the set of all even integers, and define a function $f : \mathbb{Z} \rightarrow S$ as follows:

$$f(n) = 2n \text{ for all integers } n.$$

Prove that f is one-to-one and onto.

Proof: f is one-to-one.

Suppose that $f(n_1) = f(n_2)$ for some $n_1, n_2 \in \mathbb{Z}$. Evaluating $f(n_1)$ and $f(n_2)$ gives:

$$f(n_1) = 2n_1 \quad \text{and} \quad f(n_2) = 2n_2.$$

Since $f(n_1) = f(n_2)$, we have $2n_1 = 2n_2$. Using elementary algebra, we find that $n_1 = n_2$. Therefore, f is one-to-one. ■

Proof: f is onto.

Let s be an arbitrary element of S . By the definition of S , s is an even number, so $s = 2k$ for some integer k . Evaluating $f(k)$ gives:

$$f(k) = 2k = s.$$

Since $f(k) = s$, we have shown that k is the preimage of s under f . Therefore, f is onto. ■